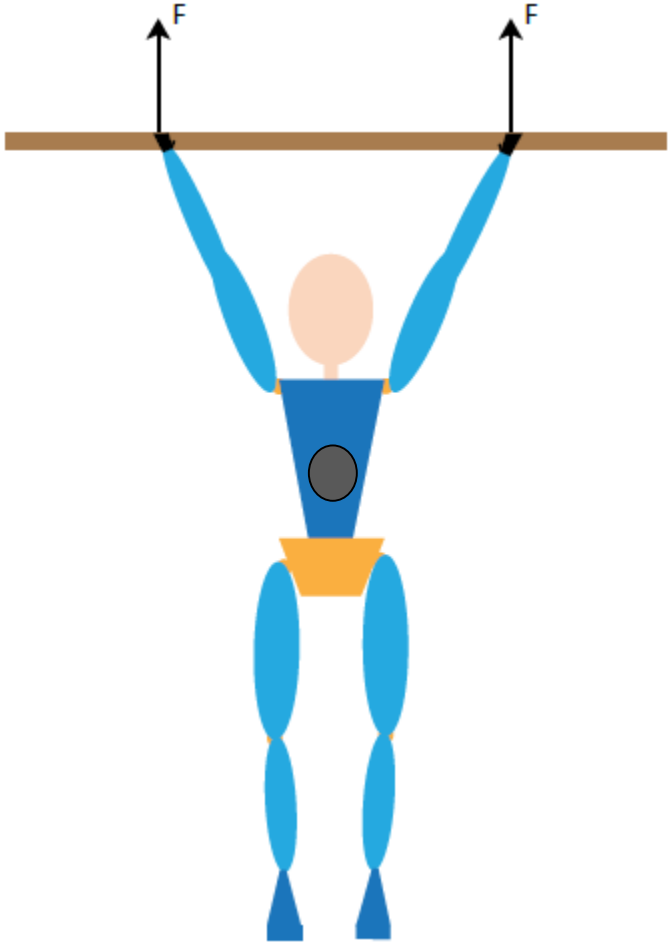
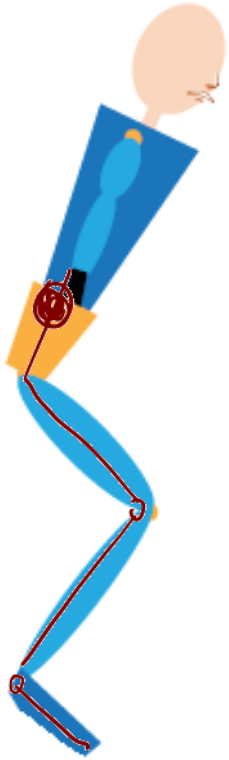
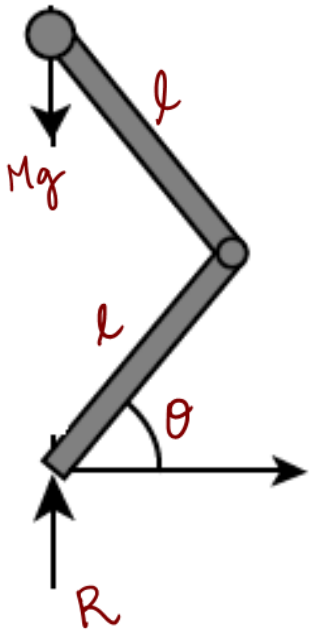
Linear motion

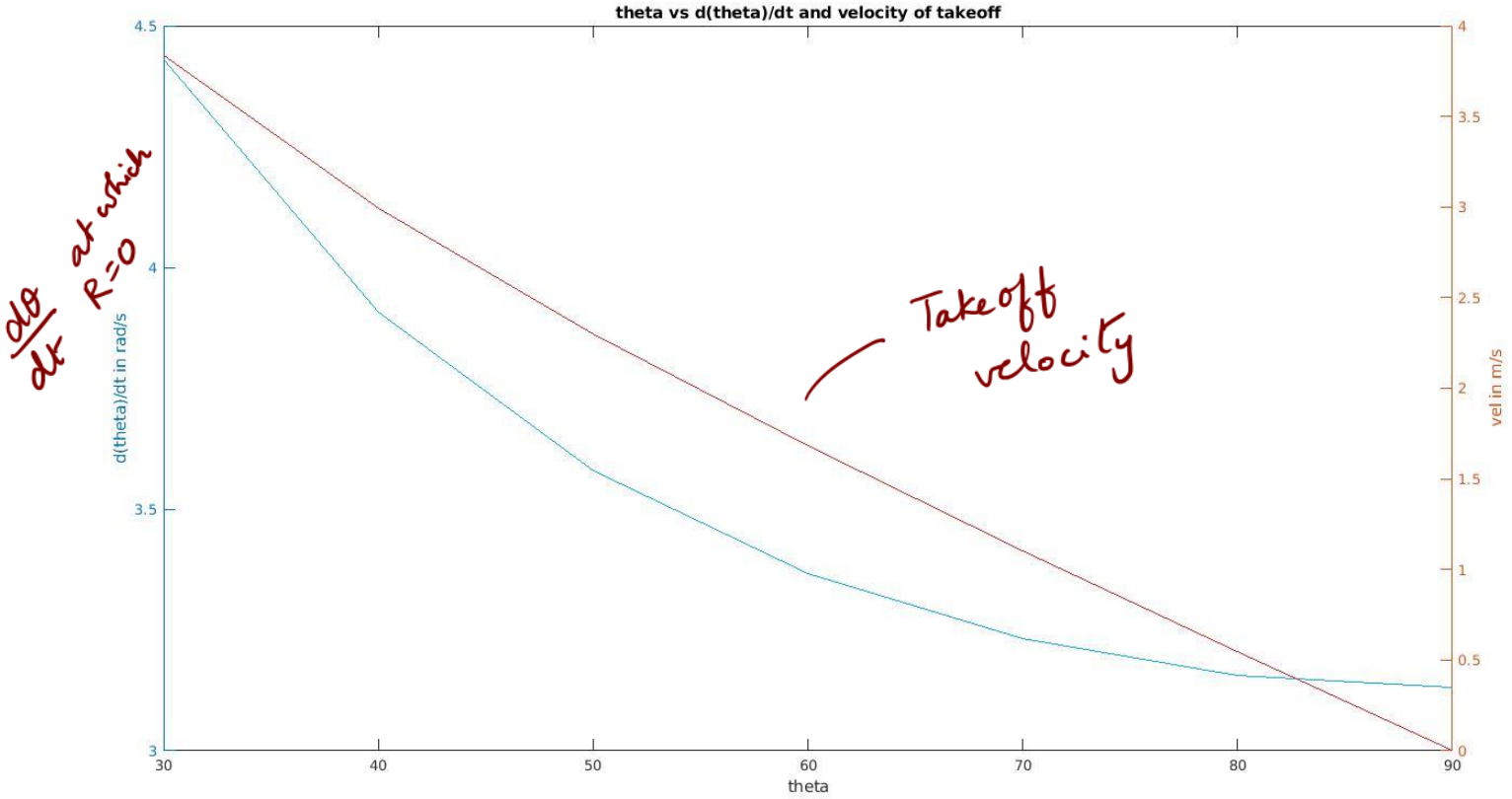


Linear motion

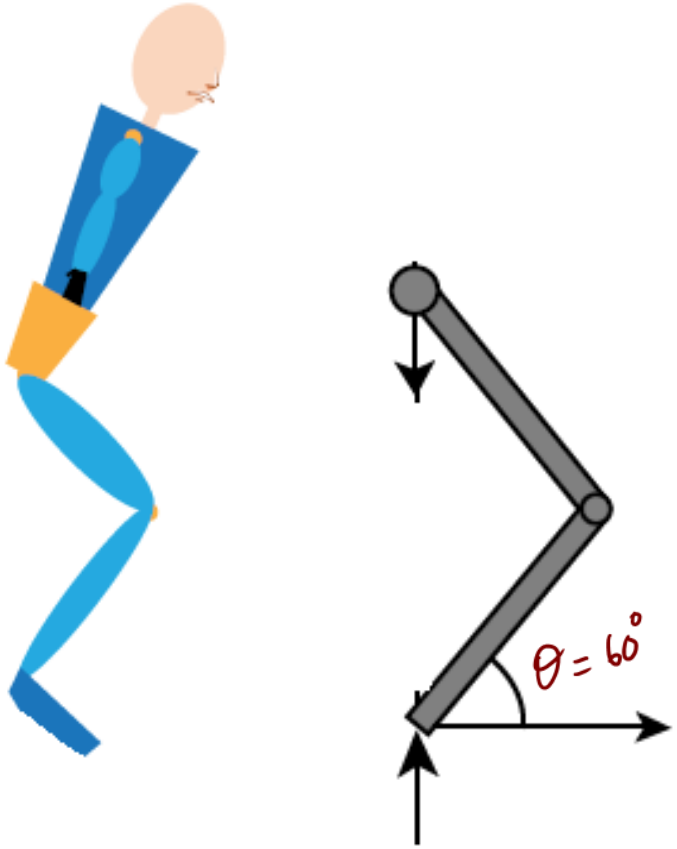
$\frac{d^2\theta}{dt^2} = 0$

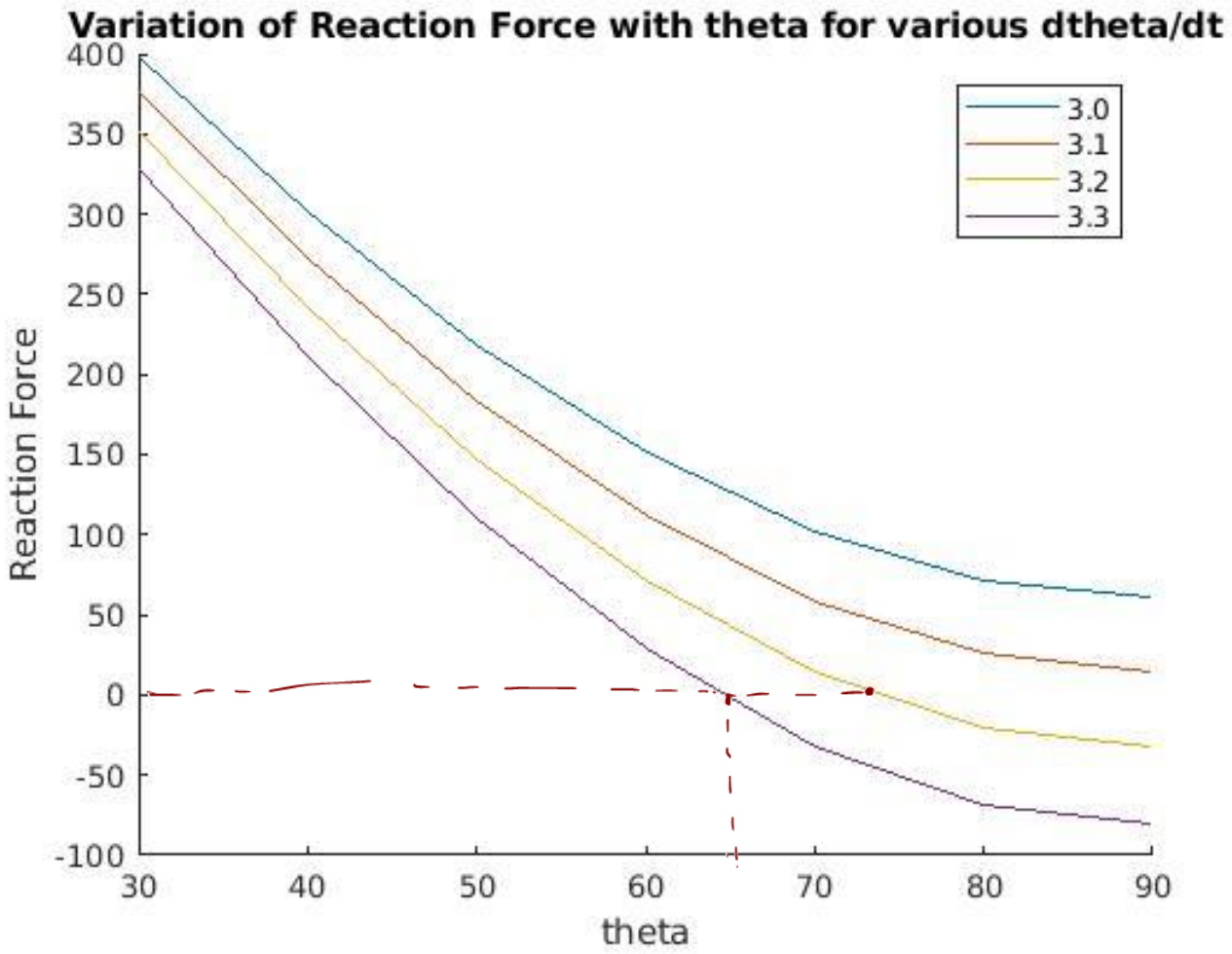






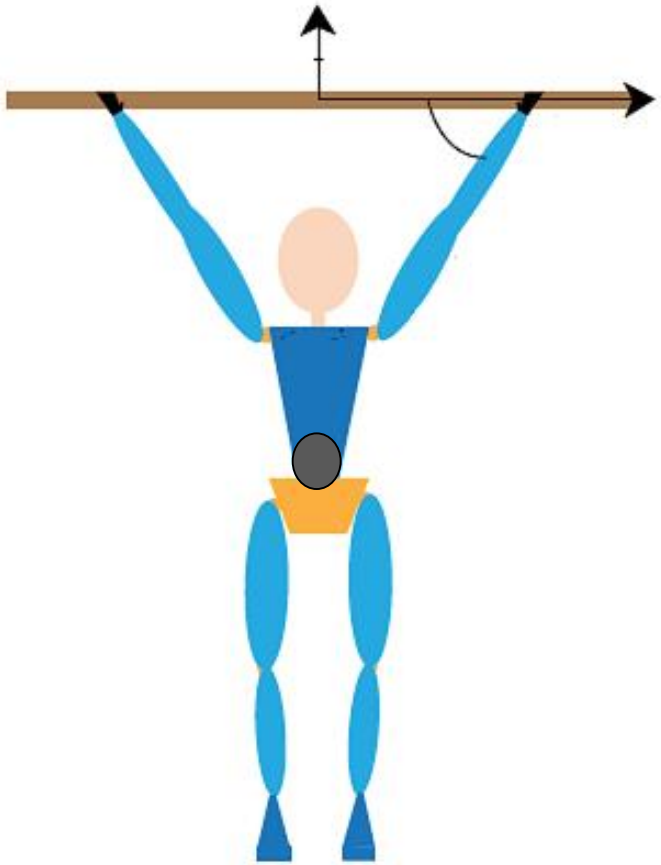
Linear motion

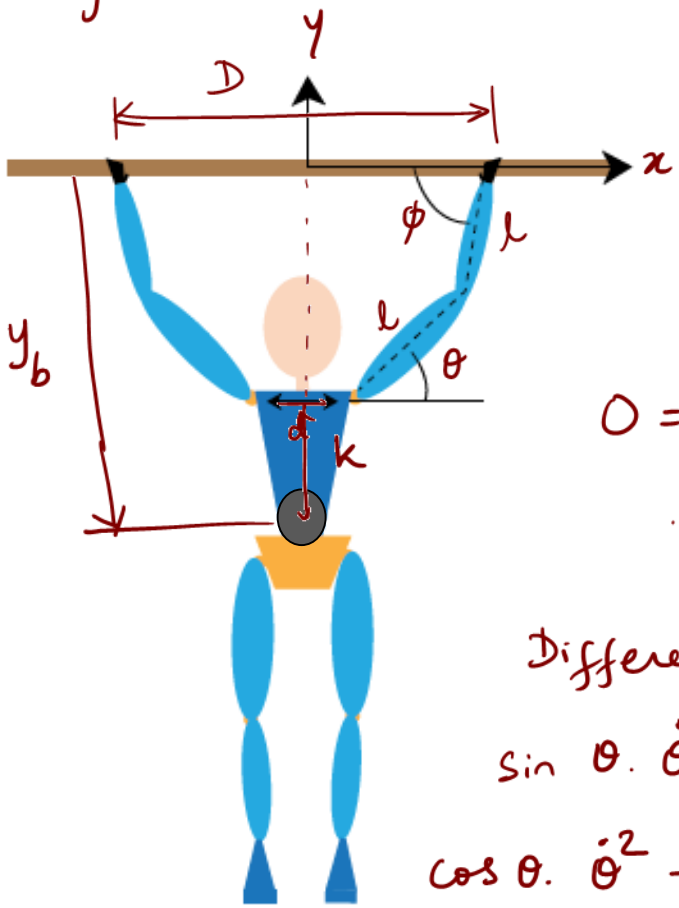




Linear motion

Assume the arms are weightless rods





$$\frac{d}{2} + l\cos\theta + l\cos\phi = \frac{D}{2}$$

$$D - d = 2(l\cos\theta + l\cos\phi)$$

constraint eqn

Differentiating w.r.t time

$$0 = -2l\sin\theta\,\dot{\theta} - 2l\sin\phi\,\dot{\phi}$$

$$\therefore \quad \dot{\theta} = -\frac{\sin\phi}{\sin\theta}\,\dot{\phi}$$

Differentiate this eqn again

$$\sin\theta\,\dot{\theta} + \sin\phi\,\dot{\phi} = 0$$

$$\cos\theta\,\dot{\theta}^2 + \sin\theta\,\ddot{\theta} + \cos\phi\,\dot{\phi}^2 + \sin\phi\,\ddot{\phi} = 0$$

$$y_b = (l\sin\phi + l\sin\theta + k)(-\hat{j})$$

$$\bar{v} = \frac{d\bar{y}}{dt} = (-l\cos\phi\,\dot{\phi} - l\cos\theta\,\dot{\theta})\,\hat{j}$$

Linear motion

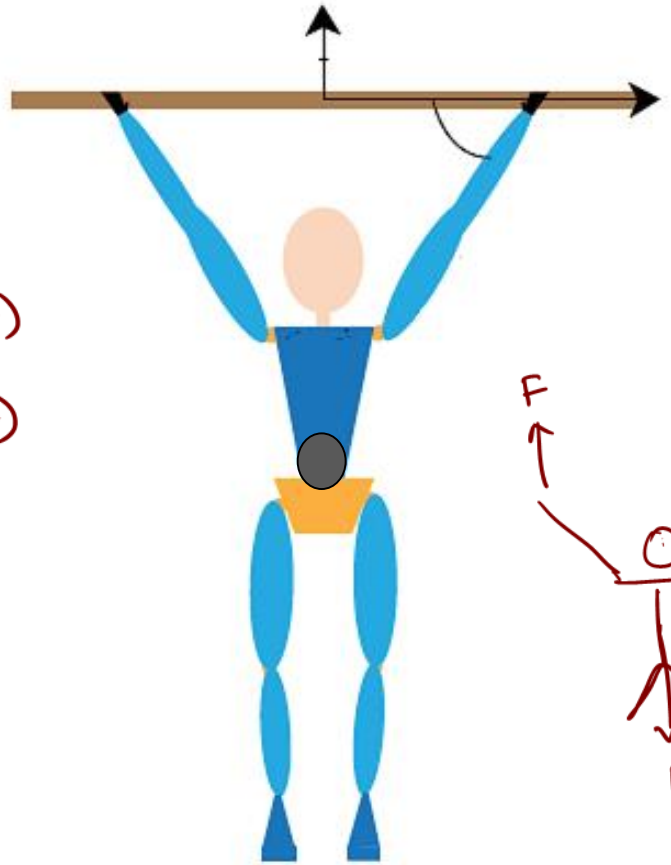
Initial conditions

$\theta = \phi = 45^\circ$, $\frac{d\theta}{dt} = 0$, $\frac{d\phi}{dt} = 0$

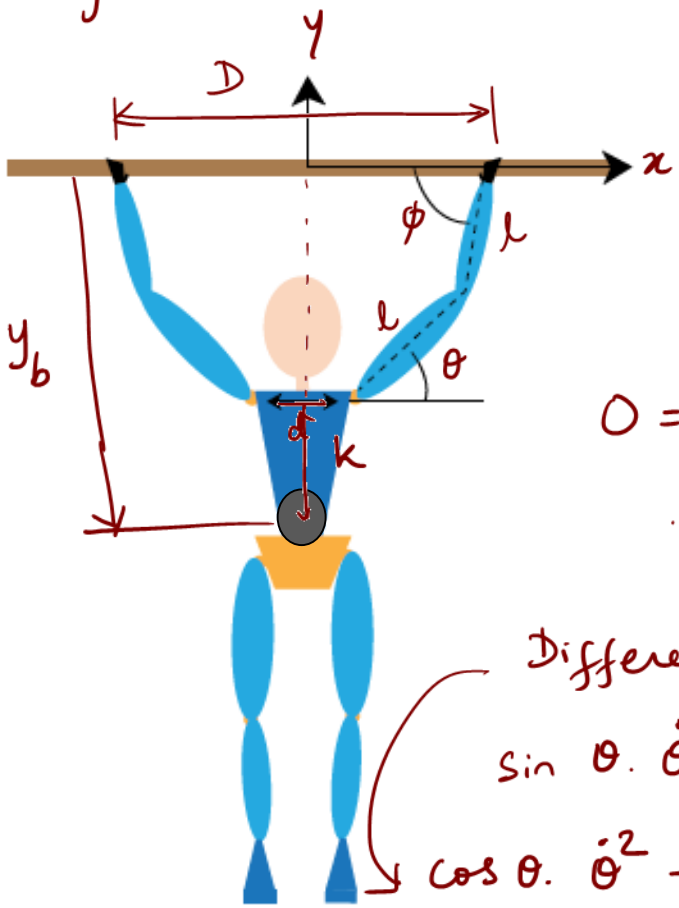
Assume $\frac{d^2\phi}{dt^2} = 2\ rad/s^2$ @ $t = 0$

Assume the arms are weightless rods

$$2F - mg = ma$$







$\theta = \theta(t)$

$\phi = \phi(t)$

Differentiating w.r.t time

$$0 = -2l \sin\theta \cdot \dot{\theta} - 2l \sin\phi \cdot \dot{\phi}$$

$$\therefore \quad \dot{\theta} = -\frac{\sin\phi}{\sin\theta}\dot{\phi} \qquad (1)$$

Differentiate this eqn (1) again

$$\sin\theta \cdot \dot{\theta} + \sin\phi\ \dot{\phi} = 0$$

$$\cos\theta \cdot \dot{\theta}^2 + \sin\theta \cdot \ddot{\theta} + \cos\phi \cdot \dot{\phi}^2 + \sin\phi \cdot \ddot{\phi} = 0$$

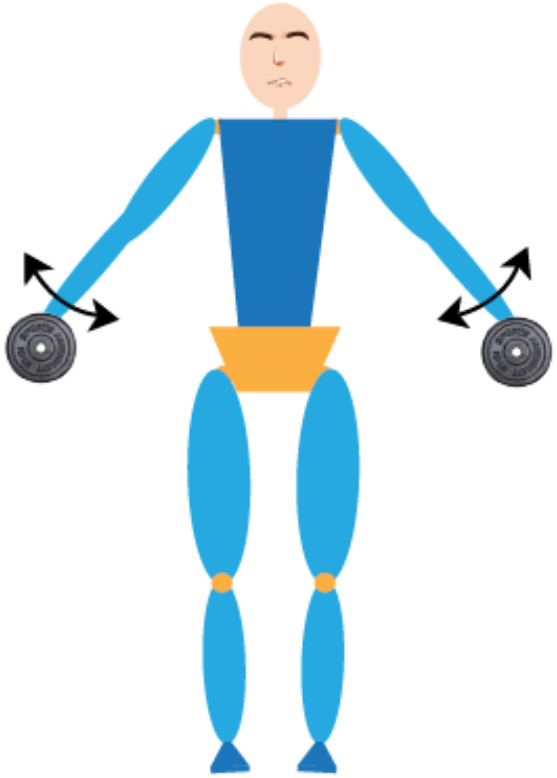
$$y_b = (l\sin\phi + l\sin\theta + k)(-\hat{j})$$

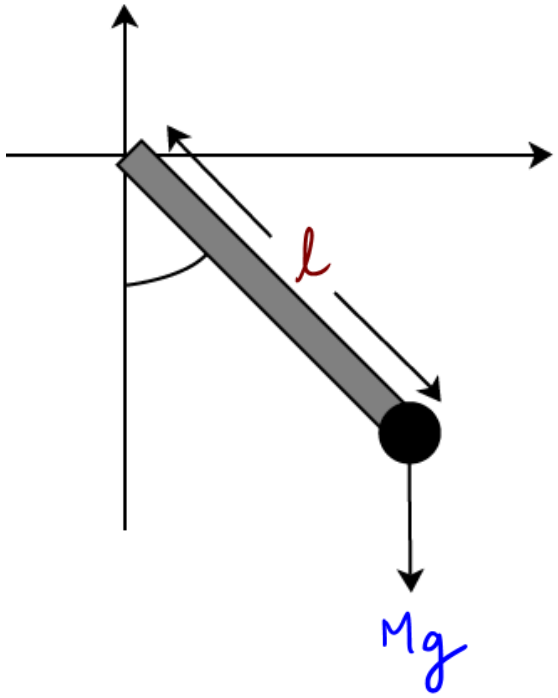
$$\bar{v} = \frac{d\bar{y}_b}{dt} = (-l\cos\phi \cdot \dot{\phi} - l\cos\theta \cdot \dot{\theta})\,\hat{j}$$

$$\bar{a} = \frac{d^2\bar{y}_b}{dt^2} = (l\sin\phi \cdot \dot{\phi}^2 - l\cos\phi \cdot \ddot{\phi} + l\sin\theta \cdot \dot{\theta}^2 - l\cos\theta \cdot \ddot{\theta})\,\hat{j}$$

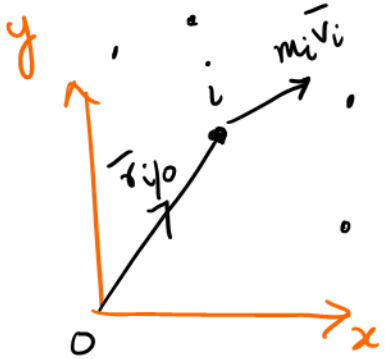
Angular motion

Arm is assumed to be weightless





Moment of momentum or angular momentum about a stationary point O, for a system of particles, is defined as



$$\bar{H}_o = \sum_{i=1}^{N} \bar{r}_{i/o} \times m_i \bar{v}_i$$

$\bar{v}_i$ is the velocity of the particle as measured in the inertial reference frame $Oxyz$

$$\frac{d\bar{H}_o}{dt} = \sum_{i=1}^{N} \frac{d\bar{r}_{i/o}}{dt} \times m_i \bar{v}_i + \sum_{i=1}^{N} \bar{r}_{i/o} \times m_i \frac{d\bar{v}_i}{dt}$$

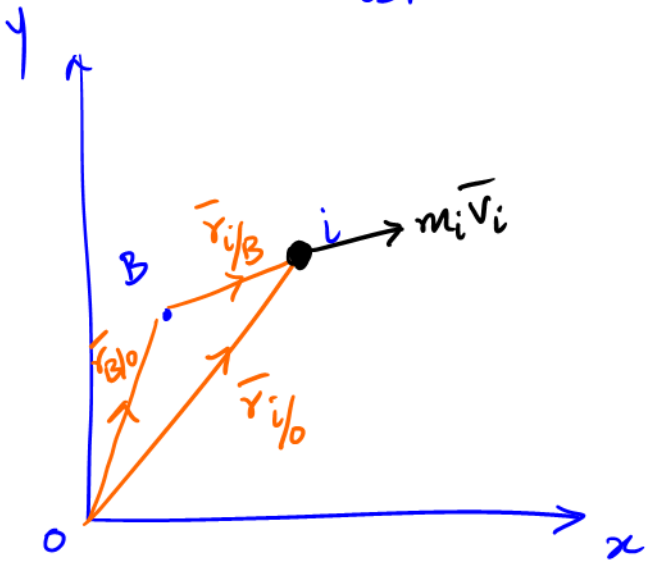
$$= \sum \underbrace{\bar{v}_i \times m_i \bar{v}_i}_{=\,0} + \sum \bar{r}_{i/o} \times m_i a_i$$

$$= \sum r_i \times \bar{F}_i$$ ~ net force on particle i

$$= \sum \bar{M}_o$$ – resultant moment about the same point of the net force on particle i & force couples

Angular momentum principle about an arbitrary point B, whose velocity w.r.t an inertial reference frame is $\bar{V}_B$. Let $\bar{L}$ be the linear momentum of the system of particles

$$\bar{L} = \sum_{i=1}^{N} m_i \bar{v}_i \qquad \bar{H}_B = \sum_{i=1}^{N} \bar{r}_{i/B} \times m_i \bar{v}_i$$



$$\bar{r}_{i/O} = \bar{r}_{B/O} + \bar{r}_{i/B}$$

$$\bar{H}_O = \sum_{i=1}^{N} \bar{r}_{i/O} \times m_i \bar{v}_i$$

$$= \sum_{i=1}^{N} \left(\bar{r}_{B/O} + \bar{r}_{i/B}\right) \times m_i \bar{v}_i$$

$$= \bar{r}_{B/O} \times \sum_{i=1}^{N} m_i \bar{v}_i + \sum_{i=1}^{N} \bar{r}_{i/B} \times m_i \bar{v}_i$$

$$= \bar{r}_{B/O} \times \bar{L} + \bar{H}_B$$

$$\bar{H}_o = \bar{r}_{B/o} \times \bar{L} + \bar{H}_B$$

Differentiating w.r.t time

$$\frac{d\bar{H}_o}{dt} = \bar{r}_{B/o} \times \frac{d\bar{L}}{dt} + \frac{d\bar{r}_{B/o}}{dt} \times \bar{L} + \frac{d\bar{H}_B}{dt} \qquad \text{——— (1)}$$

$$\bar{M}_o = \sum_{i=1}^{N} \bar{r}_{i/o} \times \bar{F}_i$$

$$= \sum_{i=1}^{N} (\bar{r}_{B/o} + \bar{r}_{i/B}) \times \bar{F}_i$$

$$= \bar{r}_{B/o} \times \sum \bar{F}_i + \sum_{i=1}^{N} \bar{r}_{i/B} \times \bar{F}_i$$

net external torque about pt. B

$$\bar{M}_o = \bar{r}_{B/o} \times \bar{F} + \bar{M}_B$$

$$\therefore \frac{d\bar{H}_o}{dt} = \bar{r}_{B/o} \times \bar{F} + \bar{M}_B \qquad \text{——— (2)}$$

Substitute (2) into (1)

$$\frac{d\bar{H}_B}{dt} + \cancel{\bar{r}_{B/O} \times \bar{F}} + \bar{v}_{B/O} \times \bar{L} = \cancel{\bar{r}_{B/O} \times \bar{F}} + \bar{M}_B$$

$$\therefore \boxed{\bar{M}_B = \frac{d\bar{H}_B}{dt} + \bar{v}_{B/O} \times \bar{L}}$$

whenever $\bar{v}_{B/O}$ is $\parallel$ to $\bar{L}$

$\bar{v}_{B/O} \times \bar{L} = 0$

Special cases: Case I : $\bar{v}_{B/O} = 0$

$$\bar{M}_B = \frac{d\bar{H}_B}{dt}$$

Case II : Suppose B is the CoM

$$\bar{L} = \sum_{i=1}^{N} m_i \bar{v}_i = \left(\sum m_i\right) \bar{v}_{B/O}$$

$\bar{v}_{B/O} \times \bar{L}$

$\therefore \bar{v}_{B/O} \times \bar{L} = \bar{v}_{B/O} \times \left(\sum m_i\right) \bar{v}_{B/O} = 0$

if B is the CoM

$$\bar{M}_B = \frac{d\bar{H}_B}{dt}$$